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Complete Backend + AI Engineering Mastery Roadmap

Goal: Master backend development fundamentals, then transition into AI engineering to build production-ready AI-powered applications.

Phase 1: Backend Foundations (Months 1-3)

1.1 Programming Language Mastery

Pick **Python** (best for AI/ML transition) or learn both Python + JavaScript

Topic	Resources
Python Fundamentals	Python Official Tutorial
Python Advanced (OOP, Decorators, Generators)	Real Python
Async Python	FastAPI Async Guide
JavaScript/Node.js (optional)	The Odin Project

Projects: - [] CLI tool for file management - [] Web scraper with error handling - [] Async data fetcher

1.2 Version Control & Git

Topic	Resources
Git Basics	Git Official Docs
GitHub Workflow	GitHub Skills
Advanced Git (rebasing, cherry-pick)	Atlassian Git Tutorials

1.3 Databases

Relational Databases

Topic	Resources
SQL Fundamentals	SQLBolt
PostgreSQL Deep Dive	PostgreSQL Tutorial
Database Design & Normalization	Database Design Course - freeCodeCamp
ORMs (SQLAlchemy)	SQLAlchemy Docs

NoSQL Databases

Topic	Resources
MongoDB	MongoDB University
Redis (Caching)	Redis University

Projects: - [] Design a normalized database schema for an e-commerce app - [] Build a caching layer with Redis

1.4 APIs & Web Frameworks

Topic	Resources
REST API Design	RESTful API Design Guide
FastAPI (Python)	FastAPI Official Docs
GraphQL Basics	How to GraphQL
API Authentication (JWT, OAuth)	Auth0 Docs

Projects: - [] Build a REST API with CRUD operations - [] Implement JWT authentication - [] Create a GraphQL API

1.5 Testing & Quality

Topic	Resources
Unit Testing (pytest)	pytest Documentation
Integration Testing	Test-Driven Development with Python
API Testing	Postman Learning Center

1.6 DevOps Fundamentals

Topic	Resources
Linux Basics	Linux Journey
Docker	Docker Curriculum
CI/CD (GitHub Actions)	GitHub Actions Docs
Cloud Basics (AWS/GCP)	AWS Free Tier Tutorials

Projects: - [] Dockerize your API - [] Set up CI/CD pipeline - [] Deploy to AWS/Railway/Render

Phase 2: Advanced Backend (Months 4-5)

2.1 System Design & Architecture

Topic	Resources
System Design Fundamentals	System Design Primer
Microservices	Microservices.io
Message Queues (RabbitMQ, Kafka)	RabbitMQ Tutorials
Event-Driven Architecture	Event-Driven Architecture Guide

2.2 Performance & Scalability

Topic	Resources
Caching Strategies	Caching Best Practices
Database Optimization	Use The Index, Luke
Load Balancing	NGINX Load Balancing
Horizontal Scaling	Scaling to Millions - High Scalability

2.3 Security

Topic	Resources
OWASP Top 10	OWASP Top 10
Secure Coding Practices	OWASP Secure Coding
API Security	API Security Best Practices

Phase 3: Mathematics for AI (Months 5-6)

Critical foundation before diving into ML/AI

3.1 Linear Algebra

Topic	Resources
Vectors, Matrices, Operations	3Blue1Brown - Essence of Linear Algebra
Eigenvalues & Eigenvectors	Khan Academy Linear Algebra
Matrix Decomposition	MIT OCW Linear Algebra

3.2 Calculus & Optimization

Topic	Resources
Derivatives & Gradients	3Blue1Brown - Essence of Calculus
Chain Rule (Backpropagation)	Calculus for ML - Coursera
Optimization (Gradient Descent)	Stanford CS229 Notes

3.3 Probability & Statistics

Topic	Resources
Probability Distributions	StatQuest YouTube
Bayes' Theorem	Seeing Theory
Statistical Inference	Think Stats (Free Book)

Phase 4: Machine Learning Foundations (Months 6-8)

4.1 Classical ML

Topic	Resources
ML Fundamentals	Google ML Crash Course
Scikit-learn	Scikit-learn Tutorials
Supervised Learning (Regression, Classification)	Andrew Ng ML Course
Unsupervised Learning (Clustering, PCA)	Stanford CS229
Model Evaluation & Validation	ML Mastery

Projects: - [] Build a classification model (spam detector) - [] Regression model (house price prediction) - [] Clustering analysis on customer data

4.2 Deep Learning

Topic	Resources
Neural Network Fundamentals	3Blue1Brown - Neural Networks
PyTorch	PyTorch Official Tutorials
CNNs (Computer Vision)	CS231n Stanford
RNNs & LSTMs (Sequences)	Colah's Blog
Transformers & Attention	The Illustrated Transformer

Projects: - [] Image classifier with PyTorch - [] Sentiment analysis model - [] Build a simple transformer from scratch

Phase 5: AI Engineering Core (Months 8-11)

This is where you become an AI Engineer, not just an ML practitioner

5.1 Large Language Models (LLMs)

Topic	Resources
How LLMs Work	Andrej Karpathy - Let's Build GPT
Prompt Engineering	OpenAI Prompt Engineering Guide

Topic	Resources
OpenAI API	OpenAI API Docs
Anthropic Claude API	Anthropic Docs
Open Source LLMs (Llama, Mistral)	Hugging Face Transformers
Fine-tuning LLMs	Hugging Face Fine-tuning Guide

Projects: - [] Build a chatbot with OpenAI API - [] Fine-tune a model on custom data - [] Compare different LLM providers

5.2 Vector Databases & Embeddings

Topic	Resources
What are Embeddings?	Vicki Boykis - Embeddings Guide
Pinecone	Pinecone Docs
Qdrant	Qdrant Documentation
Weaviate	Weaviate Docs
Chroma (Local)	Chroma Docs
pgvector (PostgreSQL)	pgvector GitHub

Projects: - [] Build semantic search over documents - [] Create a recommendation system using embeddings - [] Compare vector DB performance

5.3 Retrieval-Augmented Generation (RAG)

Topic	Resources
RAG Fundamentals	LangChain RAG Tutorial
Chunking Strategies	Pinecone Chunking Guide
Hybrid Search	Qdrant Hybrid Search
Advanced RAG (HyDE, Multi-Query)	LlamaIndex Advanced RAG
RAG Evaluation	RAGAS Framework

Projects: - [] Build a “Chat with PDF” application - [] Create a codebase Q&A system - [] Implement advanced RAG with reranking

5.4 AI Agents & Orchestration

Topic	Resources
What are AI Agents?	LangChain Agents Conceptual Guide
LangChain	LangChain Documentation
LangGraph (Recommended)	LangGraph Documentation
CrewAI	CrewAI Documentation
AutoGen (Microsoft)	AutoGen Docs
Tool Use & Function Calling	OpenAI Function Calling

Projects: - [] Build a research agent that searches and summarizes - [] Create a multi-agent system for a workflow - [] Implement human-in-the-loop agent workflows

5.5 AI Application Frameworks

Topic	Resources
LlamaIndex	LlamaIndex Docs
Vercel AI SDK	Vercel AI SDK Docs
Haystack	Haystack Documentation
Semantic Kernel (Microsoft)	Semantic Kernel Docs

Phase 6: Production AI Systems (Months 11-12)

6.1 MLOps & LLMOps

Topic	Resources
ML Pipeline Design	MLOps Guide
Model Versioning (MLflow)	MLflow Documentation
Experiment Tracking (Weights & Biases)	W&B Docs
Model Serving	BentoML
LLM Observability	LangSmith

6.2 Cost Optimization & Performance

Topic	Resources
Token Optimization	OpenAI Tokenizer
Caching LLM Responses	GPTCache
Model Quantization	Hugging Face Quantization
Batching & Streaming	vLLM

6.3 Evaluation & Testing

Topic	Resources
LLM Evaluation Metrics	Hugging Face Evaluate
Prompt Testing	Promptfoo
Red Teaming & Safety	Anthropic Safety
A/B Testing AI Features	Statsig

6.4 Deployment & Infrastructure

Topic	Resources
GPU Cloud (Modal, Replicate)	Modal Docs
Serverless AI (AWS Lambda + Bedrock)	AWS Bedrock
Edge Deployment	Ollama
Kubernetes for ML	Kubeflow

Phase 7: Specialization Tracks (Month 12+)

Choose your path based on interest:

Track A: Conversational AI

- Advanced dialogue systems
- Voice AI (Whisper, ElevenLabs)
- Multi-modal conversations

Track B: Autonomous Agents

- Complex multi-agent architectures
- Agent memory systems
- Self-improving agents

Track C: AI-Native Products

- AI UX design patterns
 - Building AI startups
 - Go-to-market for AI products
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Recommended Books

Book	Focus
Designing Machine Learning Systems	Production ML
Building LLM Apps	LLM Applications
Deep Learning	Theory (Free)
Hands-On Large Language Models	Practical LLMs

12-Month Timeline Summary

Month 1-2: Programming + Git + SQL
Month 3: APIs + FastAPI + Auth
Month 4: DevOps + Docker + Cloud
Month 5: System Design + Advanced Backend
Month 6: Math Foundations
Month 7: Classical ML
Month 8: Deep Learning + PyTorch

Month 9: LLMs + Embeddings
Month 10: RAG + Vector DBs
Month 11: AI Agents + LangGraph
Month 12: Production + MLOps
Month 12+: Specialization

Capstone Project Ideas

After completing the roadmap, build one of these:

1. **AI-Powered SaaS** - Full-stack app with AI features (like Lumienzo!)
 2. **Research Agent** - Autonomous agent that researches and writes reports
 3. **Code Assistant** - RAG over codebase + agent that can edit files
 4. **Multi-modal App** - Combine text, image, and voice AI
 5. **Open Source Tool** - Contribute to LangChain/LlamaIndex ecosystem
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Community & Staying Updated

Resource	Link
AI Engineering Newsletter	Latent Space
LangChain Discord	Join
Hugging Face Discord	Join
r/MachineLearning	Reddit
Papers With Code	paperswithcode.com
AI Twitter/X List	Follow: @karpathy, @hwchung27, @jeremyphoward

Remember: The best way to learn is to **build**. After each phase, create a project that combines everything you've learned. Your GitHub portfolio is your resume.

Roadmap created for Backend + AI Engineering mastery. Based on [roadmap.sh/backend](#) with AI engineering extensions.

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